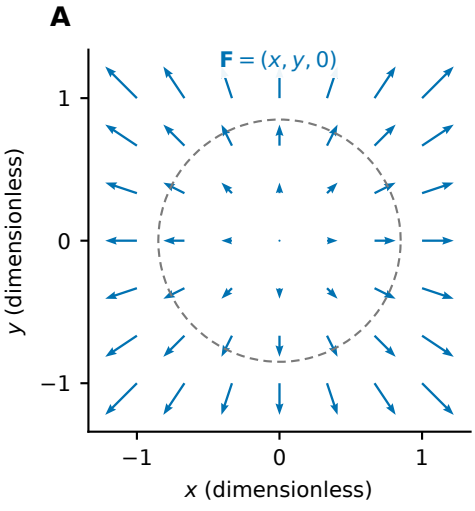
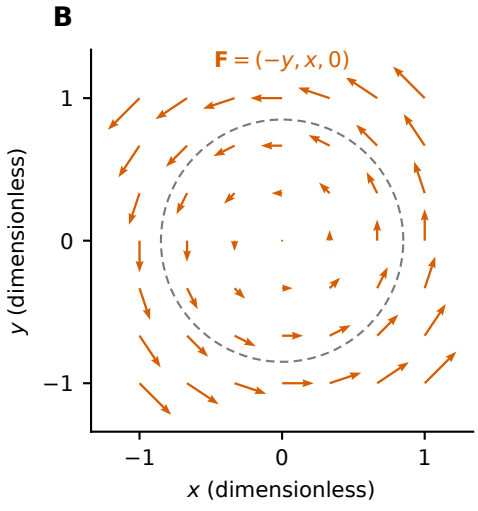


$\nabla \cdot \mathbf{F} = 2, \quad \nabla \times \mathbf{F} = 0$



$\nabla \cdot \mathbf{F} = 0, \quad \nabla \times \mathbf{F} = 2\mathbf{k}$